SUPPLEMENTARY MATERIALS: SUPPLEMENTARY MATERIALS: A JOINT-DISTRIBUTION ROUTE TO FAIR REPRESENTATIONS WITH CONTINUOUS SENSITIVE ATTRIBUTES*

YIJIN NI† AND XIAOMING HUO†

SM1. Proofs. This section collects the proofs of the results stated in the main paper, in the order in which the results appear there. The technical lemma used in the proof of Theorem 3.1 (Lemma SM2.1) is recorded in Section SM2 below.

SM1.1. Proof of Theorem 3.1 (HSIC and Expected MMD).

Proof. **Step 1: Decomposition.** By definition of the joint mean embedding and the tower property of conditional expectation,

$$\mu_{Z,S} = \mathbb{E}_{Z,S}[k_{\mathcal{Z}}(Z,\cdot) \otimes k_{\mathcal{S}}(S,\cdot)] = \mathbb{E}_S\left[\mu_{Z|S} \otimes k_{\mathcal{S}}(S,\cdot)\right],$$

and by the definition $\mu_S = \mathbb{E}_{S'}[k_{\mathcal{S}}(S',\cdot)]$ and linearity of the tensor product (with μ_Z constant in S'), $\mu_Z \otimes \mu_S = \mathbb{E}_{S'}[\mu_Z \otimes k_{\mathcal{S}}(S',\cdot)]$. Subtracting and using linearity of the Bochner integral (Lemma SM2.1),

$$\mu_{Z,S} - \mu_Z \otimes \mu_S = \mathbb{E}_S\left[(\mu_{Z|S} - \mu_Z) \otimes k_{\mathcal{S}}(S,\cdot)\right]. \tag{SM1.1}$$

Squaring the norm and writing it as an inner product gives, with $D_S := (\mu_{Z|S} - \mu_Z) \otimes k_{\mathcal{S}}(S,\cdot)$,

$$\begin{aligned}\mathrm{HSIC}(Z,S) &= \langle \mathbb{E}_S[D_S],\, \mathbb{E}_{S'}[D_{S'}]\rangle_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}}\\ &= \mathbb{E}_{S,S'}\left[\langle D_S,\, D_{S'}\rangle_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}}\right],\end{aligned}$$

where the exchange of expectation and inner product is justified by Lemma SM2.1 together with the bound $\|(\mu_{Z|S} - \mu_Z) \otimes k_{\mathcal{S}}(S,\cdot)\|_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}} \le 2\sqrt{\kappa_{\mathcal{Z}}\kappa_{\mathcal{S}}}$. By the tensor-product inner product identity $\langle a \otimes b, a' \otimes b'\rangle = \langle a, a'\rangle\langle b, b'\rangle$ and the reproducing property $\langle k_{\mathcal{S}}(S,\cdot), k_{\mathcal{S}}(S',\cdot)\rangle_{\mathcal{F}_{\mathcal{S}}} = k_{\mathcal{S}}(S,S')$,

$$\mathrm{HSIC}(Z,S) = \mathbb{E}_{S,S'}\left[\langle \mu_{Z|S} - \mu_Z, \mu_{Z|S'} - \mu_Z\rangle_{\mathcal{F}_{\mathcal{Z}}} \cdot k_{\mathcal{S}}(S,S')\right],$$

which establishes (3.3).

Step 2: Squared expected MMD bound. Starting from (SM1.1), apply the triangle inequality (Jensen's inequality for the convex norm):

$$\begin{aligned}\mathrm{HSIC}(Z,S)^{1/2} &= \left\|\mathbb{E}_S[(\mu_{Z|S} - \mu_Z) \otimes k_{\mathcal{S}}(S,\cdot)]\right\|_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}}\\ &\le \mathbb{E}_S\left[\|(\mu_{Z|S} - \mu_Z) \otimes k_{\mathcal{S}}(S,\cdot)\|_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}}\right].\end{aligned}$$

The tensor-product norm factorizes: $\|a \otimes b\|_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}} = \|a\|_{\mathcal{F}_{\mathcal{Z}}}\|b\|_{\mathcal{F}_{\mathcal{S}}}$, and $\|k_{\mathcal{S}}(S,\cdot)\|_{\mathcal{F}_{\mathcal{S}}} = \sqrt{k_{\mathcal{S}}(S,S)} \le \sqrt{\kappa_{\mathcal{S}}}$. Therefore

$$\mathrm{HSIC}(Z,S)^{1/2} \le \mathbb{E}_S\left[\|\mu_{Z|S} - \mu_Z\|_{\mathcal{F}_{\mathcal{Z}}}\sqrt{k_{\mathcal{S}}(S,S)}\right] \le \sqrt{\kappa_{\mathcal{S}}} \cdot \mathbb{E}_S[\gamma_{k_{\mathcal{Z}}}(Z \mid S, Z)].$$

Squaring both sides yields (3.4). ∎

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†H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology, Atlanta, GA 30332-0205 USA (yni64@gatech.edu, huo@gatech.edu).

SM1.2. Proof of Proposition 3.3 (HSIC controls kernel-smoothed conditional deviations).

Proof. Boundedness of $k_{\mathcal{Z}}, k_{\mathcal{S}}$ guarantees that the mean embeddings μ_Z, μ_S exist as Bochner integrals and that the centered random elements $k_{\mathcal{S}}(S,\cdot) - \mu_S \in \mathcal{F}_{\mathcal{S}}$, $k_{\mathcal{Z}}(Z,\cdot) - \mu_Z \in \mathcal{F}_{\mathcal{Z}}$ have finite second moments, so the tensor-product expectation $C_{SZ} = \mathbb{E}[(k_{\mathcal{S}}(S,\cdot) - \mu_S) \otimes (k_{\mathcal{Z}}(Z,\cdot) - \mu_Z)]$ defines a Hilbert–Schmidt operator $\mathcal{F}_{\mathcal{Z}} \to \mathcal{F}_{\mathcal{S}}$ with $\|C_{SZ}\|_{\mathrm{HS}}^2 = \mathrm{HSIC}(Z,S)$ (Gretton et al., 2005). By Cauchy–Schwarz on Hilbert–Schmidt operators applied to f,

$$\|C_{SZ} f\|_{\mathcal{F}_{\mathcal{S}}}^2 \;\le\; \|C_{SZ}\|_{\mathrm{op}}^2 \, \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2 \;\le\; \|C_{SZ}\|_{\mathrm{HS}}^2 \, \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2 \;=\; \mathrm{HSIC}(Z,S) \, \|f\|_{\mathcal{F}_{\mathcal{Z}}}^2,$$

where $\|A\|_{\mathrm{op}} \le \|A\|_{\mathrm{HS}}$ holds for any compact operator on a Hilbert space. The reformulation in terms of m_f uses the identity $C_{SZ} f = \mathbb{E}_S[m_f(S)\, k_{\mathcal{S}}(S,\cdot)]$, which follows by writing

$$\begin{aligned}
(C_{SZ} f)(s) &= \langle k_{\mathcal{S}}(s,\cdot),\, C_{SZ} f\rangle_{\mathcal{F}_{\mathcal{S}}} = \mathbb{E}\big[\langle k_{\mathcal{S}}(s,\cdot), k_{\mathcal{S}}(S,\cdot) - \mu_S\rangle_{\mathcal{F}_{\mathcal{S}}} \, \langle k_{\mathcal{Z}}(Z,\cdot) - \mu_Z,\, f\rangle_{\mathcal{F}_{\mathcal{Z}}}\big] \\
&= \mathbb{E}\big[(k_{\mathcal{S}}(s,S) - \mu_S(s))\,(f(Z) - \mathbb{E}[f(Z)])\big] = \mathbb{E}_S\big[m_f(S)(k_{\mathcal{S}}(s,S) - \mu_S(s))\big],
\end{aligned}$$

and noting that the $\mu_S(s)$ term contributes $\mathbb{E}_S[m_f(S)]\,\mu_S(s) = 0$ since m_f has mean zero (cf. Fukumizu et al., 2007, Lemma 5). ∎

SM1.3. Proof of Theorem 3.4 (Empirical conditional-gap control by HSIC, projected version) and Corollary 3.5.

Proof of Theorem 3.4. We work with empirical quantities throughout. Denote by $\hat{d}_{S_i} := k_{\mathcal{Z}}(Z_i,\cdot) - \hat{\mu}_Z$ the empirical centered embedding at $S = S_i$, with $\hat{\mu}_Z = n^{-1}\sum_{j=1}^n k_{\mathcal{Z}}(Z_j,\cdot)$, and write $\boldsymbol{\delta}_f := (\hat{\delta}_{f,1}, \ldots, \hat{\delta}_{f,n})^\top$ for the centered prediction-deviation vector.

Step 1: Spectral identity for $\widehat{\mathrm{HSIC}}$ using positive eigenpairs only. Let $K \in \mathbb{R}^{n\times n}$ have entries $K_{ij} = k_{\mathcal{Z}}(Z_i, Z_j)$ and $\widetilde{K} = HKH$. The biased V-statistic estimator is

$$\widehat{\mathrm{HSIC}}(Z,S) = \frac{1}{n^2}\mathrm{tr}(KHLH) = \frac{1}{n^2}\mathrm{tr}(\widetilde{K}\,\widetilde{L}).$$

Since L is symmetric positive semi-definite and $H = H^\top$ with $H^2 = H$, $\widetilde{L} = HLH$ is symmetric PSD. Let $\hat{\lambda}_1 \ge \cdots \ge \hat{\lambda}_r > 0$ be the positive eigenvalues of $n^{-1}\widetilde{L}$ with associated orthonormal eigenvectors $\hat{e}_1, \ldots, \hat{e}_r \in \mathbb{R}^n$, where $r = \mathrm{rank}(\widetilde{L})$. Then

$$\frac{1}{n}\widetilde{L} = \sum_{j=1}^r \hat{\lambda}_j\, \hat{e}_j \hat{e}_j^\top, \qquad P_{\widetilde{L}} = \sum_{j=1}^r \hat{e}_j \hat{e}_j^\top.$$

Substituting and using $\widehat{\mathrm{HSIC}} = n^{-1}\mathrm{tr}(\widetilde{K} \cdot n^{-1}\widetilde{L})$,

$$\widehat{\mathrm{HSIC}}(Z,S) = \frac{1}{n}\sum_{j=1}^r \hat{\lambda}_j \cdot \hat{e}_j^\top \widetilde{K} \hat{e}_j. \tag{SM1.2}$$

Step 2: Lower bound via the smallest positive eigenvalue. Since $\widetilde{K} = HKH$ is PSD and $\hat{\lambda}_S = \min_{1\le j\le r} \hat{\lambda}_j > 0$, every term in (SM1.2) satisfies $\hat{\lambda}_j \cdot \hat{e}_j^\top \widetilde{K} \hat{e}_j \ge \hat{\lambda}_S \cdot \hat{e}_j^\top \widetilde{K} \hat{e}_j$. Therefore

$$\widehat{\mathrm{HSIC}}(Z,S) \;\ge\; \frac{\hat{\lambda}_S}{n}\sum_{j=1}^r \hat{e}_j^\top \widetilde{K} \hat{e}_j \;=\; \frac{\hat{\lambda}_S}{n}\,\mathrm{tr}\big(P_{\widetilde{L}}\, \widetilde{K}\, P_{\widetilde{L}}\big), \tag{SM1.3}$$

where we used $\sum_{j=1}^{r} \hat{e}_j^\top \widetilde{K} \hat{e}_j = \operatorname{tr}(P_{\widetilde{L}} \widetilde{K} P_{\widetilde{L}})$ (since $P_{\widetilde{L}} = \sum_{j=1}^{r} \hat{e}_j \hat{e}_j^\top$, $P_{\widetilde{L}}^2 = P_{\widetilde{L}}$, and the cyclic-trace identity).

Step 3: Bound the projected prediction deviation by $\operatorname{tr}(P_{\widetilde{L}} \widetilde{K} P_{\widetilde{L}})$**.** By the reproducing property, $\hat{\delta}_{f,i} = \langle f, \hat{d}_{S_i} \rangle_{\mathcal{F}_\mathcal{Z}}$, so the vector $\boldsymbol{\delta}_f$ has entries $(\boldsymbol{\delta}_f)_i = \langle f, \hat{d}_{S_i} \rangle_{\mathcal{F}_\mathcal{Z}}$. For any unit vector $u \in \mathbb{R}^n$, the linear combination $\sum_i u_i \hat{d}_{S_i} \in \mathcal{F}_\mathcal{Z}$ and $u^\top \boldsymbol{\delta}_f = \langle f, \sum_i u_i \hat{d}_{S_i} \rangle_{\mathcal{F}_\mathcal{Z}}$. By Cauchy–Schwarz in $\mathcal{F}_\mathcal{Z}$,

$$(u^\top \boldsymbol{\delta}_f)^2 \;\le\; \|f\|_{\mathcal{F}_\mathcal{Z}}^2 \Big\| \sum_i u_i \hat{d}_{S_i} \Big\|_{\mathcal{F}_\mathcal{Z}}^2 \;=\; \|f\|_{\mathcal{F}_\mathcal{Z}}^2 \, u^\top \widetilde{K} \, u,$$

where the last equality uses $\langle \hat{d}_{S_i}, \hat{d}_{S_j} \rangle_{\mathcal{F}_\mathcal{Z}} = \widetilde{K}_{ij}$. Applying this with $u = \hat{e}_j$ for $j = 1, \ldots, r$ and summing,

$$\sum_{j=1}^{r} (\hat{e}_j^\top \boldsymbol{\delta}_f)^2 \;\le\; \|f\|_{\mathcal{F}_\mathcal{Z}}^2 \sum_{j=1}^{r} \hat{e}_j^\top \widetilde{K} \hat{e}_j \;=\; \|f\|_{\mathcal{F}_\mathcal{Z}}^2 \operatorname{tr}\big(P_{\widetilde{L}} \widetilde{K} P_{\widetilde{L}}\big).$$

The left-hand side equals $\|P_{\widetilde{L}} \boldsymbol{\delta}_f\|_2^2 = \sum_{i=1}^{n} (P_{\widetilde{L}} \boldsymbol{\delta}_f)_i^2$ since $\{\hat{e}_j\}_{j=1}^{r}$ is an orthonormal basis for the column span of $\widetilde{L}$. Dividing by n,

$$\text{(SM1.4)} \qquad \frac{1}{n} \sum_{i=1}^{n} (P_{\widetilde{L}} \boldsymbol{\delta}_f)_i^2 \;\le\; \frac{\|f\|_{\mathcal{F}_\mathcal{Z}}^2}{n} \operatorname{tr}\big(P_{\widetilde{L}} \widetilde{K} P_{\widetilde{L}}\big).$$

Step 4: Combine. Combining (SM1.3) (which gives $\operatorname{tr}(P_{\widetilde{L}} \widetilde{K} P_{\widetilde{L}}) \le n\, \widehat{\mathrm{HSIC}} / \hat{\lambda}_S$) with (SM1.4),

$$\frac{1}{n} \sum_{i=1}^{n} (P_{\widetilde{L}} \boldsymbol{\delta}_f)_i^2 \;\le\; \frac{\|f\|_{\mathcal{F}_\mathcal{Z}}^2}{\hat{\lambda}_S} \widehat{\mathrm{HSIC}}(Z, S),$$

which is (3.6). □

Proof of Corollary 3.5. If $\operatorname{rank}(\widetilde{L}) = n - 1$, then the null space of $\widetilde{L}$ is one-dimensional and contains $\mathbf{1}$ (since $\widetilde{L}\mathbf{1} = HLH\mathbf{1} = 0$), so it equals $\operatorname{span}\{\mathbf{1}\}$ and $P_{\widetilde{L}}$ is the orthogonal projection onto its complement. The centered vector $\boldsymbol{\delta}_f$ satisfies $\mathbf{1}^\top \boldsymbol{\delta}_f = \sum_i (f(Z_i) - \bar{f}_n) = 0$, hence lies in the column span of $\widetilde{L}$, so $P_{\widetilde{L}} \boldsymbol{\delta}_f = \boldsymbol{\delta}_f$ and $\sum_i (P_{\widetilde{L}} \boldsymbol{\delta}_f)_i^2 = \sum_i \hat{\delta}_{f,i}^2$. The first inequality of the corollary then follows from (3.6). The second (absolute-gap) inequality is Jensen applied to $x \mapsto x^2$:

$$\hat{\Delta}_f^2 = \Big(\frac{1}{n} \sum_i |\hat{\delta}_{f,i}| \Big)^2 \le \frac{1}{n} \sum_i \hat{\delta}_{f,i}^2 \le \frac{\|f\|_{\mathcal{F}_\mathcal{Z}}^2}{\hat{\lambda}_S} \widehat{\mathrm{HSIC}}(Z, S).$$

□

Population analog. The empirical bound above does not require any spectral-gap hypothesis on the population kernel operator. For completeness we record a population analog under an additional hypothesis: a strict spectral gap $\lambda_2 > 0$ for the centered kernel operator $T_{\tilde{k}_\mathcal{S}}$. This hypothesis fails for Gaussian kernels on continuous $\mathcal{S}$ (where $\inf_{j \ge 1} \lambda_j = 0$). Under that extra hypothesis, the key step is the spectral inequality $\mathrm{HSIC}(Z, S) \ge \lambda_2 \cdot \mathbb{E}_S[\|d_S\|_{\mathcal{F}_\mathcal{Z}}^2]$, and we elaborate on three technical points.

Why the null space contributes zero. The centered kernel $\tilde{k}_\mathcal{S}$ has the constant function $e_0 \equiv 1$ in its null space ($\lambda_0 = 0$). For characteristic kernels, this is the *only* null eigenfunction on $L^2(P_S)$ (Fukumizu et al., 2009). The contribution of e_0 to the

Parseval expansion of $\mathbb{E}_S[\|d_S\|^2_{\mathcal{F}_\mathcal{Z}}]$ is $\|\mathbb{E}_S[d_S \cdot 1]\|^2_{\mathcal{F}_\mathcal{Z}} = \|\mathbb{E}_S[d_S]\|^2_{\mathcal{F}_\mathcal{Z}} = 0$, so the null space does not affect the bound.

Why each $\lambda_j > 0$. For a characteristic kernel $k_\mathcal{S}$ and a distribution P_S with full support on $\mathcal{S}$, the integral operator $T_{k_\mathcal{S}}$ is injective on $L^2(P_S)$ (Fukumizu et al., 2009). The centered operator $T_{\tilde{k}_\mathcal{S}}$ inherits this property on the zero-mean subspace, ensuring $\lambda_j > 0$ for all $j \geq 1$. Note that for compact operators on infinite-dimensional spaces (e.g., Gaussian kernels on continuous $\mathcal{S}$), the eigenvalues $\lambda_j \to 0$ as $j \to \infty$, so $\inf_{j\geq 1} \lambda_j = 0$; the spectral gap assumption $\lambda_2 > 0$ used here is therefore an additional hypothesis that holds for finite-rank approximations but not at the population level for standard kernels. The empirical theorem above avoids this issue entirely by working with the strictly positive $\hat{\lambda}_S$.

The operator inequality. It remains to show that $\mathbb{E}_S[\|d_S\|^2_{\mathcal{F}_\mathcal{Z}}] \leq \frac{1}{\lambda_2} \cdot \mathrm{HSIC}(Z, S)$. From the Mercer expansion of $\tilde{k}_\mathcal{S}$ and Parseval's identity applied to $s \mapsto d_s$, and using that $c_0 = 0$ (as shown above):

$$\mathrm{HSIC}(Z, S) = \sum_{j\geq 1} \lambda_j c_j \geq \lambda_2 \sum_{j\geq 1} c_j = \lambda_2 \cdot \mathbb{E}_S[\|d_S\|^2_{\mathcal{F}_\mathcal{Z}}],$$

where $c_j := \|\mathbb{E}_S[d_S \cdot e_j(S)]\|^2_{\mathcal{F}_\mathcal{Z}} \geq 0$. This proves the population analog under the additional hypothesis $\lambda_2 > 0$.

SM1.4. Proof of Proposition 3.9 (Population-level equivalence of HSIC and EIPM).

Proof. ($\Rightarrow$) If $\mathrm{HSIC}(Z, S) = 0$, then $\mu_{Z,S} = \mu_Z \otimes \mu_S$ in $\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}$. Because $k_\mathcal{Z}$ and $k_\mathcal{S}$ are characteristic, the product kernel $k_\mathcal{Z} \otimes k_\mathcal{S}$ is $\mathcal{I}$-characteristic, i.e., its joint mean embedding distinguishes $P_{Z,S}$ from $P_Z \otimes P_S$ (Theorem 3 of Szabó and Sriperumbudur, 2018, taking $M = 2$). Hence $\mu_{Z,S} = \mu_Z \otimes \mu_S$ implies $P_{Z,S} = P_Z \otimes P_S$, i.e., $Z \perp S$. Hence $P_{Z|S=s} = P_Z$ for P_S-almost every s, so $\mathrm{MMD}(P_{Z|S=s}, P_Z) = 0$ a.s., and $\mathrm{EIPM}_\mathcal{V}(Z; S) = \mathbb{E}_S[\mathrm{MMD}(P_{Z|S}, P_Z)] = 0$.

($\Leftarrow$) Suppose $\mathrm{EIPM}_\mathcal{V}(Z; S) = 0$. Since $\mathrm{MMD}(P_{Z|S=s}, P_Z) \geq 0$, this implies $\mathrm{MMD}(P_{Z|S=s}, P_Z) = 0$ for P_S-almost every s. Because $k_\mathcal{Z}$ is characteristic, MMD is a metric on probability distributions, so $P_{Z|S=s} = P_Z$ a.s., i.e., $Z \perp S$. Independence then implies that the joint mean embedding factorizes, $\mu_{Z,S} = \mu_Z \otimes \mu_S$, so $\mathrm{HSIC}(Z, S) = \|\mu_{Z,S} - \mu_Z \otimes \mu_S\|^2_{\mathcal{F}_{\mathcal{Z}\otimes\mathcal{S}}} = 0$. ∎

SM1.5. Proof of Theorem 3.13 (Subgroup-fairness equivalence).

Proof. We show $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (iv) \Rightarrow (i)$.

$(i) \Rightarrow (ii)$. This is the same argument as in Proposition 3.9: under characteristic $k_\mathcal{Z}, k_\mathcal{S}$, the product kernel $k_\mathcal{Z} \otimes k_\mathcal{S}$ is $\mathcal{I}$-characteristic (Szabó and Sriperumbudur 2018, Theorem 3, $M = 2$), so $\mu_{Z,S} = \mu_Z \otimes \mu_S$ implies $P_{Z,S} = P_Z \otimes P_S$.

$(ii) \Rightarrow (iii)$. On the Polish product space $\mathcal{Z} \times \mathcal{S}$, the disintegration theorem (Kallenberg, 2002, Thm. 5.4) yields a regular conditional probability $\{P_{Z|S=s}\}_{s\in\mathcal{S}}$ such that for every Borel $A \subseteq \mathcal{Z}, B \subseteq \mathcal{S}$,

$$P_{Z,S}(A \times B) = \int_B P_{Z|S=s}(A)\, dP_S(s).$$

Under (ii), the same Borel set $A \times B$ also satisfies $P_{Z,S}(A \times B) = P_Z(A)\, P_S(B) = \int_B P_Z(A)\, dP_S(s)$. Comparing the two expressions, $\int_B [P_{Z|S=s}(A) - P_Z(A)]\, dP_S(s) = 0$ for every Borel B, so $P_{Z|S=s}(A) = P_Z(A)$ for P_S-a.e. s. Choose a countable algebra $\mathcal{A}$ generating the Borel σ-algebra of $\mathcal{Z}$ (which exists since $\mathcal{Z}$ is Polish); the exceptional

P_S-null sets, taken over the countable family $\mathcal{A}$, have countable union of P_S-measure zero. Hence $P_{Z|S=s}|_{\mathcal{A}} = P_Z|_{\mathcal{A}}$ for P_S-a.e. s, and by the π-λ theorem $P_{Z|S=s} = P_Z$ on the full Borel σ-algebra for P_S-a.e. s. Equivalence of "some" and "every" regular conditional version follows from the P_S-a.e. uniqueness clause of the disintegration theorem.

$(iii) \Rightarrow (iv)$. For every bounded measurable f and P_S-a.e. s,

$$\mathbb{E}[f(Z) \mid S = s] = \int f(z)\, dP_{Z|S=s}(z) = \int f(z)\, dP_Z(z) = \mathbb{E}[f(Z)],$$

where the second equality uses (iii).

$(iv) \Rightarrow (i)$. Since $\mathcal{Z}$ is Polish, fix a countable dense subset $\{z'_k\}_{k\geq 1} \subseteq \mathcal{Z}$. Apply (iv) with $f(\cdot) = k_{\mathcal{Z}}(\cdot, z'_k)$ for each k (bounded by $\sqrt{\kappa_{\mathcal{Z}}}$ since $k_{\mathcal{Z}}$ is bounded). For each k, condition (iv) holds outside a P_S-null set N_k. The countable union $N := \bigcup_k N_k$ is also P_S-null. For every $s \notin N$ and every k,

$$\mu_{Z|S=s}(z'_k) = \mathbb{E}[k_{\mathcal{Z}}(Z, z'_k) \mid S = s] = \mathbb{E}[k_{\mathcal{Z}}(Z, z'_k)] = \mu_Z(z'_k).$$

Both $z' \mapsto \mu_{Z|S=s}(z')$ and $z' \mapsto \mu_Z(z')$ are continuous on $\mathcal{Z}$ (continuity of the kernel $k_{\mathcal{Z}}$ in its first argument, which holds for the bounded characteristic kernels considered here), so equality on the dense subset $\{z'_k\}$ extends to equality on all of $\mathcal{Z}$. Hence $\mu_{Z|S=s} = \mu_Z$ in $\mathcal{F}_{\mathcal{Z}}$ for P_S-a.e. s, which gives $\mathrm{MMD}(P_{Z|S=s}, P_Z) = \|\mu_{Z|S=s} - \mu_Z\|_{\mathcal{F}_{\mathcal{Z}}} = 0$ for P_S-a.e. s. Therefore $\mathbb{E}_S[\mathrm{MMD}(P_{Z|S}, P_Z)] = 0$, and by Theorem 3.1 (population-level upper bound), $\mathrm{HSIC}(Z, S) \leq \kappa_{\mathcal{S}} \cdot (\mathbb{E}_S[\mathrm{MMD}(P_{Z|S}, P_Z)])^2 = 0$. ∎

SM1.6. Proof of Corollary 3.17 (HSIC vs. EIPM efficiency).

Proof. The HSIC rate is Proposition 3.15. The EIPM rate is Theorem 4 of Kong et al. (2025), which establishes

$$\epsilon_n = \gamma_n^2 + \frac{\log n}{\sqrt{n\gamma_n}} \Big(1 + \log \mathcal{N}\big(\sqrt{\gamma_n/n}, \mathcal{V}, \|\cdot\|_\infty\big)\Big)^{1/2},$$

optimized at $\gamma_n^{\mathrm{opt}} \propto n^{-1/5}$, yielding $\epsilon_n^{\mathrm{opt}} = O((\log n) \cdot n^{-2/5})$. Since $n^{-1/2}$ is asymptotically faster than $n^{-2/5}$, the HSIC estimator therefore has a faster nominal rate than the EIPM estimator. We emphasize that these are estimation rates for two different population functionals; the comparison is at the estimator level, not a claim that HSIC estimates a common target more accurately than EIPM. ∎

SM2. Additional Notation and Technical Lemmas.

LEMMA SM2.1 (Fubini's theorem for RKHS-valued integrals). *Let k be a bounded kernel, i.e., $\sup_{x\in\mathcal{X}} k(x, x) \leq \kappa < \infty$. Then for any random variable X on $\mathcal{X}$, the mean embedding $\mu_X = \mathbb{E}[k(X, \cdot)]$ exists as a Bochner integral in $\mathcal{F}$, and the exchange of expectation and inner product in the proof of Theorem 3.1 is justified.*

Proof. By the reproducing property,

$$\|k(x, \cdot)\|_{\mathcal{F}} = \sqrt{k(x, x)} \leq \sqrt{\kappa}, \quad \text{so} \quad \mathbb{E}[\|k(X, \cdot)\|_{\mathcal{F}}] \leq \sqrt{\kappa} < \infty,$$

guaranteeing existence of the Bochner integral. The exchange of expectation and inner product follows from continuity of the inner product together with Fubini's theorem. ∎

SM3. Regularized empirical kHGR. This appendix records a regularized empirical bound linking $\widehat{\mathrm{HSIC}}$ to a Tikhonov-regularized empirical kHGR. The unregularized empirical kHGR requires inverting empirical covariance operators with arbitrarily small positive eigenvalues, so a meaningful finite-sample statement is naturally regularized; see Fukumizu et al. (2007) for the kCCA interpretation. The FRHSIC guarantees in the main text do not depend on this result.

Empirical covariance operators. Define the empirical centered cross-covariance operator $\hat{\Sigma}_{ZS} : \mathcal{F}_{\mathcal{S}} \to \mathcal{F}_{\mathcal{Z}}$ by

$$\hat{\Sigma}_{ZS} = \frac{1}{n}\sum_{i=1}^{n}\big(k_{\mathcal{Z}}(Z_i,\cdot) - \hat{\mu}_Z\big) \otimes \big(k_{\mathcal{S}}(S_i,\cdot) - \hat{\mu}_S\big),$$

where $\hat{\mu}_Z, \hat{\mu}_S$ are the empirical mean embeddings, and define $\hat{\Sigma}_{ZZ}, \hat{\Sigma}_{SS}$ analogously. All three operators are finite-rank (rank at most n), hence Hilbert–Schmidt without further assumptions on the kernels. By a direct expansion, $\widehat{\mathrm{HSIC}}(Z,S) = \|\hat{\Sigma}_{ZS}\|_{\mathrm{HS}}^2$.

Regularized empirical kHGR. For ridge parameters $\eta_Z, \eta_S > 0$, define the Tikhonov-regularized empirical kHGR

$$\widehat{\mathrm{kHGR}}_{\eta_Z,\eta_S}(Z,S) \;:=\; \big\|(\hat{\Sigma}_{ZZ} + \eta_Z I)^{-1/2}\,\hat{\Sigma}_{ZS}\,(\hat{\Sigma}_{SS} + \eta_S I)^{-1/2}\big\|_{\mathrm{op}}.$$

PROPOSITION SM3.1 (Regularized empirical sandwich bound). *Let $k_{\mathcal{Z}}, k_{\mathcal{S}}$ be characteristic kernels and let $\eta_Z, \eta_S > 0$. Then*

$$\widehat{\mathrm{kHGR}}_{\eta_Z,\eta_S}(Z,S)^2 \;\le\; \frac{\widehat{\mathrm{HSIC}}(Z,S)}{\eta_Z\,\eta_S}. \tag{SM3.1}$$

Proof. Set $A = (\hat{\Sigma}_{ZZ}+\eta_Z I)^{-1/2}$, $B = \hat{\Sigma}_{ZS}$, $C = (\hat{\Sigma}_{SS}+\eta_S I)^{-1/2}$. Each operator $\hat{\Sigma}_{ZZ}, \hat{\Sigma}_{SS}$ is positive semi-definite, so $\hat{\Sigma}_{ZZ} + \eta_Z I \succeq \eta_Z I$ and $\hat{\Sigma}_{SS} + \eta_S I \succeq \eta_S I$. Consequently

$$\|A\|_{\mathrm{op}} \;\le\; 1/\sqrt{\eta_Z}, \qquad \|C\|_{\mathrm{op}} \;\le\; 1/\sqrt{\eta_S}.$$

By the operator-norm $\le$ Hilbert–Schmidt-norm inequality and the standard submultiplicative bound $\|ABC\|_{\mathrm{op}} \le \|A\|_{\mathrm{op}}\,\|B\|_{\mathrm{HS}}\,\|C\|_{\mathrm{op}}$ (valid because the Hilbert–Schmidt norm dominates the operator norm),

$$\widehat{\mathrm{kHGR}}_{\eta_Z,\eta_S}(Z,S) = \|ABC\|_{\mathrm{op}} \;\le\; \|A\|_{\mathrm{op}}\,\|B\|_{\mathrm{HS}}\,\|C\|_{\mathrm{op}} \;\le\; \frac{\|\hat{\Sigma}_{ZS}\|_{\mathrm{HS}}}{\sqrt{\eta_Z\,\eta_S}}.$$

Squaring and using $\widehat{\mathrm{HSIC}}(Z,S) = \|\hat{\Sigma}_{ZS}\|_{\mathrm{HS}}^2$ gives (SM3.1). ∎

The proposition makes no claim about the unregularized limit $\eta_Z, \eta_S \downarrow 0$, for which the spectrum of the empirical covariance operators must be controlled separately. Population-level zero-equivalence of HSIC and kHGR under characteristic kernels is independent of this regularization choice.

SM4. Experimental Details.

SM4.1. Datasets.

Adult Income. The UCI Adult dataset contains $n \approx 30{,}000$ records (after removing missing values). Features include numeric variables and one-hot encoded categoricals ($d = 107$ after encoding). The target is whether income exceeds \$50K/year. The sensitive attribute is age (continuous, range 17–90). All features and the sensitive attribute are min-max scaled to $[0, 1]$.

ACS Income. The ACS Income dataset (Ding et al., 2021) is drawn from the 2018 American Community Survey (California) via the folktables package. We subsample n = 20,000 individuals. The target is whether income exceeds \$50K/year. The sensitive attribute is age (continuous, range 17–94). Features include 9 demographic and employment variables.

MEPS. The Medical Expenditure Panel Survey (Panel 19, 2015) (Romano et al., 2020) contains $n \approx 13{,}000$ respondents after removing records with missing values (negative sentinel codes). The target is healthcare utilization ≥ 10 visits (binary classification). The sensitive attribute is age (continuous, range 17–85). Features include 19 variables covering demographics, health conditions, insurance status, and income.

Communities & Crime. The UCI Communities & Crime dataset contains $n \approx 2{,}000$ communities. The target is violent crimes per population (regression). The sensitive attribute is racial composition (continuous). Columns with >20% missing values are dropped; remaining missing values are imputed with column medians.

COMPAS. The ProPublica COMPAS dataset contains $n \approx 6{,}000$ defendants after standard filtering (Angwin et al., 2016). The target is two-year recidivism (binary classification). The sensitive attribute is age (continuous). Features include prior counts, charge degree (one-hot), sex, and race (one-hot), yielding $d = 16$ features.

SM4.2. Model Architecture. Following Kong et al. (2025), the encoder h is a 2-layer MLP: $d_X \to 50 \to 50$ with SELU activations (hidden dimension $H = 50$, output representation dimension $Z = 50$). The prediction head f is a linear layer: $50 \to 1$ (no hidden layers). This architecture matches FREM exactly, ensuring a fair comparison. The adversary in LAFTR and ADV uses the same 2-layer structure ($50 \to 50 \to n_{\text{out}}$) with SELU activation.

SM4.3. Training Details. All models are trained for 200 epochs with the Adam optimizer (learning rate 10^{-3}, default $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay 0), batch size 256, and the standard PyTorch default initialization. For FRHSIC, kernel bandwidths are set via the median heuristic: $\sigma_{\mathcal{S}}$ is computed once on the training data, and $\sigma_{\mathcal{Z}}$ is recomputed every 20 epochs on the current mini-batch of encoded representations. For baselines (FREM, Reg-GDP, etc.), we use fixed $\sigma_Z = 1.0$ for the Z-kernel after min-max scaling to $[0, 1]$, following Kong et al. (2025); the additional S-smoothing bandwidths used by FREM ($\gamma = 0.5$) and Reg-GDP (Nadaraya–Watson bandwidth 0.2) are taken from their source publications and described per-method below. We do not modify per-method bandwidth conventions, ensuring each method is evaluated as published.

Train/validation/test splits. For all real datasets we use an 80%/20% random train/test split. Hyperparameter selection (i.e., choosing λ along the Pareto frontier) is performed using the same training split as the model fit; for each method and each dataset we sweep λ over the full grid and report results across the resulting Pareto frontier (no separate validation split is used to pick a single λ, since the goal of the comparison is to characterize the fairness–accuracy tradeoff curve, not to choose one operating point). We repeat 5 times with different random splits and report mean $\pm$ standard deviation on the held-out test set.

Regularization-strength (λ) grids. For all methods on all real datasets, we sweep

$$\lambda \in \{0.1,\ 1.0,\ 10.0,\ 100.0,\ 500.0\}.$$

For the Equal Opportunity and multi-sensitive-attribute experiments (Section SM5) the grid is $\lambda \in \{0.1, 1.0, 10.0, 50.0\}$.

SM4.4. Baseline Implementations and Hyperparameter Grids.

Reg-GDP. We implement the GDP regularizer from Jiang et al. (2022) using kernel-smoothed conditional expectations on a grid of 30 evenly-spaced points over the range of S. Reg-GDP uses two distinct bandwidths: a Gaussian Z-kernel bandwidth $\sigma_Z = 1.0$ (matching the FREM convention of Kong et al. (2025) on min-max-scaled features) and a Nadaraya–Watson smoothing bandwidth 0.2 on S for the conditional expectation; both follow the values reported by Jiang et al. (2022).

FREM. We implement the weighted EIPM estimator from Kong et al. (2025) using the MMD discriminator. Kernel-smoothed weights are computed with a Gaussian kernel on S with bandwidth $\gamma = 0.5$ (following Kong et al. (2025)). For computational tractability we subsample $n_{\text{anchor}} = 32$ anchor points per batch for the EIPM computation; see Section SM4.5 below for details.

LAFTR. We bin the continuous sensitive attribute into 4 quartiles and apply the adversarial FRL method of Madras et al. (2018). The adversary is trained with alternating gradient steps (one adversary step per encoder step). The adversary is a 2-layer MLP ($50 \to 50 \to 4$) with SELU activations.

ADV. The continuous-target adversary of Grari et al. (2022), with the same 2-layer architecture as LAFTR but with a single scalar output (predicting S); trained with one adversary step per encoder step.

MMD-binned. Bins continuous S into $n_{\text{bins}} = 10$ equal-width bins and applies pairwise MMD across bins; bin centers serve as the smoothing grid.

dCor. We use the unbiased empirical distance correlation of Székely et al. (2007) as the fairness penalty; no further hyperparameters beyond λ.

SM4.5. FREM 32-anchor subsampling. The FREM EIPM objective of Kong et al. (2025) computes, at each mini-batch, an integral over s of a weighted MMD term:

$$\widehat{\text{EIPM}}_n(Z; S) = \int_{\mathcal{S}} \widehat{\text{MMD}}^2\big(\widehat{P}_{Z|S=s},\, \widehat{P}_Z\big)\, d\widehat{P}_S(s).$$

Estimating this expectation by Monte Carlo with all n training points as anchors would require $O(n^3)$ work per batch (an $n \times n$ kernel matrix evaluated at each of n anchors). For computational tractability we instead draw $n_{\text{anchor}} = 32$ uniformly-random anchor points per mini-batch and average the MMD terms over those anchors. With batch size 256 this is a $32/256 = 12.5\%$ Monte Carlo subsample and reduces the per-batch cost to $O(n_{\text{anchor}} \cdot n^2)$. The number 32 was chosen to match the FREM implementation in our codebase (see `experiments/real_data.py`, function `train_frem`); we report results with $n_{\text{anchor}} = 32$ throughout.

FREM anchor-subsampling sensitivity (limitation). The 32-anchor subsampling we use for FREM is a computational approximation; an anchor-count sweep is the natural next step to characterize the sensitivity of FREM's reported Pareto frontier to the anchor budget, and we plan to include such a sweep in a follow-up version of this work. The per-epoch wall-clock cost grows linearly with the anchor count, so the principal cost of the sweep is compute rather than implementation.

Baseline bandwidth sensitivity (limitation). The bandwidths reported in Section SM4 (FRHSIC median heuristic, FREM $\gamma = 0.5$ and $\sigma_Z = 1.0$, Reg-GDP Nadaraya–Watson bandwidth 0.2 and $\sigma_Z = 1.0$) are the values reported in the source publications. We retain published bandwidths in this paper to avoid favoring any particular method by retuning, but a full bandwidth-sensitivity sweep for the FREM and Reg-GDP baselines is the natural next step, and we plan to include such a sweep in a follow-up version of this work.

Paired significance test (limitation). We report mean and standard deviation across $n = 5$ random splits in Table 2 of the main paper but do not run a formal paired test (e.g., Wilcoxon signed-rank), since 5 paired observations is too few for reliable inference at conventional significance levels. Increasing the number of repeats to enable a credible paired comparison is the natural next step and we plan to include such an analysis in a follow-up version of this work.

SM4.6. Reproducibility checklist. We summarize the elements needed to reproduce the experiments end to end.

- **Random seed.** A single base seed `SEED = 42` is set at the top of each experiment script for `numpy`, `torch`, and the train/test split RNG. The 5 repeats use seeds $\texttt{SEED}, \texttt{SEED} + 1, \ldots, \texttt{SEED} + 4$. See `experiments/real_data.py:27` and `experiments/utils.py`.
- **Train/validation/test split.** 80%/20% random train/test split; no separate validation split (rationale above). Repeated 5 times with different splits.
- **Preprocessing.** Min–max scaling for the features and the sensitive attribute is fit on the *training split only* and then applied to the held-out test split (`experiments/real_data.py`, `run_single_dataset`). The Adult and Crime cells of Table 2 in the main paper were re-run under this leakage-free scaling protocol; the resulting numbers agree with our earlier leakage-affected numbers to within standard deviations on essentially every cell, with the most visible shifts in the MMD and ADV rows of Crime (where the small sample size $n \approx 2000$ amplifies any scaling difference). The remaining cells (ACS Income, MEPS, COMPAS) still report the earlier leakage-affected numbers and will be updated in the camera-ready revision. Categoricals are one-hot encoded; rows with missing values are dropped except in Communities & Crime, where columns with >20% missing are dropped and remaining missing values are imputed with column medians.
- **Model architecture.** Encoder h: 2-layer MLP $d_X \to 50 \to 50$ with SELU activations. Predictor f: linear $50 \to 1$. Adversaries (LAFTR / ADV / MMD-binned): 2-layer MLP $50 \to 50 \to n_{\text{out}}$ with SELU.
- **Optimizer.** Adam, learning rate 10^{-3}, $\beta_1 = 0.9, \beta_2 = 0.999$, $\varepsilon = 10^{-8}$, weight decay 0. PyTorch defaults.
- **Batch size and epochs.** Batch size 256, 200 epochs.
- **Kernel bandwidths.** FRHSIC: median heuristic for $\sigma_{\mathcal{S}}$ (fixed at start of training, on training data) and for $\sigma_{\mathcal{Z}}$ (recomputed every 20 epochs on the current mini-batch). Baselines: fixed $\sigma_Z = 1.0$ on min-max-scaled Z, FREM $\gamma = 0.5$, Reg-GDP NW bandwidth 0.2.
- **λ grids.** Main experiments: $\lambda \in \{0.1, 1.0, 10.0, 100.0, 500.0\}$. EO and multi-sensitive: $\lambda \in \{0.1, 1.0, 10.0, 50.0\}$.
- **Baseline hyperparameter grids.** See Section SM4, “Baseline Implementations”; all baselines use the same encoder, predictor, optimizer, batch size, and number of epochs as FRHSIC, varying only the per-method fairness-penalty hyperparameters listed above. The adversary step ratio (LAFTR, ADV) is fixed at 1:1.
- **FREM anchor count.** $n_{\text{anchor}} = 32$ uniformly-sampled anchors per mini-batch; see Section SM4.5.
- **Hardware.** All experiments were run on a single workstation with one NVIDIA RTX-class GPU; the experiment code falls back to CPU if no CUDA device is available (see the device selector in `experiments/utils.py`). Run-

time numbers in Section 4 of the main paper are wall-clock per-epoch on this single device, averaged over 5 epochs after a warm-up epoch. Training the full λ-sweep across all methods and datasets fits within $\sim$24 GPU-hours.

- **Software.** Python 3.11, PyTorch 2.x, NumPy 1.26, scikit-learn 1.3, folktables (for ACS Income). See `experiments/README.md` for the exact versions used at submission time.
- **Code availability.** All scripts to reproduce the figures and tables are in the `experiments/` directory of the supplementary code archive; each main-paper figure/table is generated by a single named script (e.g. `main_table.py`, `pareto_curves.py`, `runtime.py`, `convergence.py`).

SM4.7. Pareto summary: lowest GDP within 1% of the unfair-baseline performance. For a single-number summary of the fairness–accuracy tradeoff referenced in Section 4 of the main paper, Table SM1 reports, for each (method, dataset) pair, the operating point on the λ-sweep that attains the *lowest GDP* subject to the constraint that the prediction performance is within 1% of the unfair baseline (Acc $\geq 0.99 \cdot \mathrm{Acc}_{\mathrm{Unfair}}$ for classification; MSE $\leq 1.01 \cdot \mathrm{MSE}_{\mathrm{Unfair}}$ for regression). Entries are read off the Pareto sweep underlying Figure 3 of the main paper. "—" denotes that no λ in our grid satisfies the constraint for that method on that dataset. Values are means across 5 random splits.

Table SM1

Pareto summary: (Acc / MSE, GDP) at the lowest-GDP operating point with prediction performance within 1% of the unfair baseline. Adult/ACS/MEPS/COMPAS report Acc; Crime reports MSE $\times 10^2$. *"—" indicates no λ in the sweep satisfies the constraint. Read alongside Figure 3 of the main paper, since a tight 1% band can exclude operating points that are visually close on the frontier.*

Method	Adult	ACS Income	MEPS	Crime	COMPAS
Unfair (ref.)	(.848, .790)	(.789, .482)	(.804, .587)	(1.84, .028)	(.649, .117)
FRHSIC (Ours)	(.846, .545)	(.792, .416)	(.802, .507)	(1.82, .029)	(.653, .105)
FREM	(.847, .173)	(.787, .262)	(.799, .303)	(1.81, .028)	(.653, .059)
Reg-GDP	(.844, .092)	(.782, .113)	(.800, .363)	(1.85, .030)	(.644, .056)
ADV	(.847, .542)	(.784, .257)	(.804, .479)	—	(.642, .065)
MMD (binned)	(.850, .257)	(.786, .142)	(.801, .393)	(1.85, .027)	(.648, .082)
LAFTR (binned)	(.838, .394)	(.785, .303)	(.796, .262)	(1.88, .027)	(.644, .089)
dCor	(.845, .205)	(.789, .262)	(.802, .477)	(1.79, .028)	(.643, .078)

Caveats. The 1% accuracy band is intentionally tight, and on some datasets (notably COMPAS, where multiple methods cluster near the unfair-baseline accuracy) the band excludes operating points that are visually close on the Pareto frontier (Figure 3). Reg-GDP can drive GDP to near zero on Adult and ACS, but only at accuracy losses well outside the 1% band; those points are visible on Figure 3 but do not appear in the table above. The single-number summary should therefore be read alongside the full Pareto curves rather than as a standalone ranking.

SM5. Additional Experimental Results.

SM5.1. Equal Opportunity. We evaluate the EO extension (Section 3.6 of the main paper) on Adult and COMPAS, comparing FRHSIC-DP (HSIC on all samples) to FRHSIC-EO (HSIC on $Y = 1$ subset only). Table SM2 reports accuracy and EO-GDP (GDP restricted to the $Y = 1$ subset) across λ values. On COMPAS, the DP variant achieves comparable or better EO-GDP than the EO variant at similar accuracy across all λ values. On Adult, the EO variant achieves slightly lower EO-GDP at high λ (0.352 vs. 0.418 at $\lambda = 50$), but the DP variant simultaneously reduces full GDP from 0.889 to 0.240—a broader fairness guarantee. Overall, enforcing full demographic parity via HSIC provides a strong baseline for Equal Opportunity, with the dedicated EO variant offering marginal gains on larger datasets.

Table SM2

Equal Opportunity results: FRHSIC-DP vs. FRHSIC-EO. EO-GDP is GDP restricted to $Y = 1$ samples. FRHSIC-DP achieves comparable EO fairness while also enforcing DP.

Dataset	λ	Method	Acc	EO-GDP (↓)
Adult	0.1	DP	0.851	0.453
		EO	0.849	0.406
	1.0	DP	0.846	0.403
		EO	0.850	0.416
	10.0	DP	0.846	0.437
		EO	0.850	0.392
	50.0	DP	0.840	0.418
		EO	0.849	0.352
COMPAS	0.1	DP	0.659	0.067
		EO	0.662	0.070
	1.0	DP	0.662	0.070
		EO	0.662	0.075
	10.0	DP	0.657	0.064
		EO	0.655	0.065
	50.0	DP	0.599	0.033
		EO	0.659	0.076

SM5.2. Multiple Sensitive Attributes. We test FRHSIC with two continuous sensitive attributes (age and hours-per-week) on Adult. Table SM3 compares two approaches: (1) a product kernel on joint (S_1, S_2) (FRHSIC-Joint), and (2) a sum of per-attribute HSIC terms (FRHSIC-Sum). Both approaches reduce GDP with respect to each attribute as λ increases while preserving accuracy, confirming that HSIC naturally extends to the multi-attribute setting without modification. The Joint approach achieves slightly lower GDP at $\lambda = 50$ (Age: 0.306 vs. 0.678; Hours: 0.517 vs. 0.759), likely because the product kernel captures cross-attribute structure. Figure SM1 visualizes the Pareto frontiers.

SM6. Additional Theoretical Augmentations. This section collects two augmentations referenced from the main paper: a necessity-for-counterfactual-fairness result and a uniform-concentration bound that bridges training-sample HSIC to its population counterpart.

SM6.1. Necessity for Counterfactual Fairness. The joint-distribution route to independence also clarifies the relationship between HSIC-fairness and the causal-fairness literature.

Proposition SM6.1 (HSIC-fairness as a necessary condition for counterfactual fairness). *Assume the SCM in Section 2.8 of the main paper. Suppose the encoder h is a deterministic function so that $Z = h(X) = h(f_X(S, U_X))$. If the prediction $\widehat{Y} = f(Z)$ is counterfactually fair (Section 2.8 of the main paper) for every f in a*

TABLE SM3

Multiple sensitive attributes on Adult (age + hours-per-week). GDP is reported separately for each attribute. Both product-kernel (Joint) and sum-of-HSIC (Sum) approaches reduce GDP for both attributes as λ increases.

Method (λ)	Acc	GDP(Age) ($\downarrow$)	GDP(Hours) ($\downarrow$)
Unfair	0.843	0.837	1.142
FRHSIC-Joint ($\lambda = 1$)	0.844	0.779	1.029
FRHSIC-Joint ($\lambda = 10$)	0.846	0.553	0.857
FRHSIC-Joint ($\lambda = 50$)	0.851	0.306	0.517
FRHSIC-Sum ($\lambda = 1$)	0.847	0.753	1.055
FRHSIC-Sum ($\lambda = 10$)	0.848	0.485	0.826
FRHSIC-Sum ($\lambda = 50$)	0.849	0.678	0.759

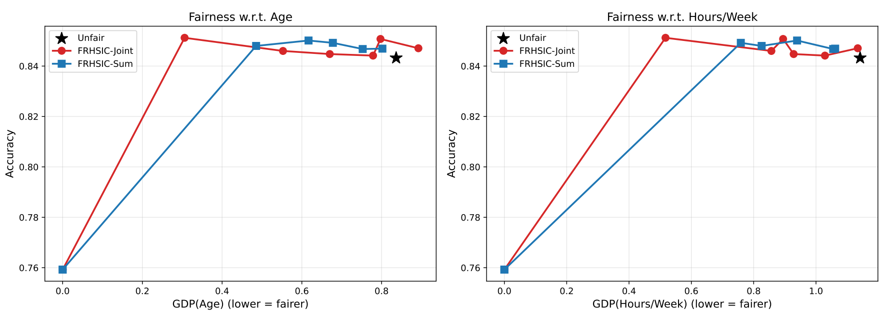

FIG. SM1. *Multiple sensitive attributes: fairness–accuracy Pareto frontiers for FRHSIC-Joint (product kernel) and FRHSIC-Sum (sum of per-attribute HSIC), evaluated on Adult with age and hours-per-week as sensitive attributes.*

class $\mathcal{F}$ rich enough to identify equality of distributions in $\mathcal{Z}$ (e.g., $\mathcal{F}$ is the unit ball of an RKHS with characteristic kernel $k_{\mathcal{Z}}$), then $\mathrm{HSIC}(Z, S) = 0$.

Proof. Counterfactual fairness for every $f \in \mathcal{F}$ implies, in particular, that for all $s' \in \mathcal{S}$ in the support of S and all $f \in \mathcal{F}$,

$$\mathbb{E}[f(Z_{S \leftarrow s'})] = \mathbb{E}[f(Z)],$$

where $Z_{S \leftarrow s'} = h(f_X(s', U_X))$. Since $S = U_S$ and U_X are independent under the SCM, and the marginal of $Z_{S \leftarrow s'}$ over U_X has the same distribution as the marginal of Z given $S = s'$ (because intervening with $S = s'$ matches the conditional given $S = s'$ when S is exogenous (Pearl, 2009, Section 3.2.2)), we obtain

$$\mathbb{E}[f(Z) \mid S = s'] = \mathbb{E}[f(Z)] \quad \text{for } P_S\text{-a.e. } s'.$$

By Theorem 3.13 of the main paper, this is equivalent to $\mathrm{HSIC}(Z, S) = 0$. ∎

In words, if a representation can support counterfactually fair predictions for every (sufficiently rich) prediction head, the representation must already satisfy the observational-independence condition that HSIC enforces. The converse does not hold: $\mathrm{HSIC}(Z, S) = 0$ ensures that the marginal distribution of $f(Z)$ does not depend on S, but counterfactual fairness is a strictly pointwise statement about individual-level interventions, which can fail when the structural dependence on S is observationally invisible (e.g., $Z = S \oplus U_X$ with binary S, U_X uniform yields $Z \perp S$ marginally but $Z_{S\leftarrow s'} \neq Z_{S\leftarrow s}$ pointwise). HSIC-fairness should therefore be understood as the strongest *observational* fairness consequence of counterfactual fairness, not as a substitute for it.

SM6.2. Uniform Concentration over the Encoder Class. The empirical Theorem 3.4 of the main paper controls the projected empirical second-moment conditional gap $\frac{1}{n}\sum_i (P_{\widetilde{L}}\boldsymbol{\delta}_f)_i^2$ through $\widehat{\mathrm{HSIC}}(h(X), S)$ for a single encoder h (and, under the full-rank hypothesis of Corollary 3.5, the unprojected $\frac{1}{n}\sum_i \hat{\delta}_{f,i}^2$). In practice, h is learned from the same data, so the bound must hold uniformly over a hypothesis class $\mathcal{H}$ of encoders. We invoke Corollary 23 of Ni and Huo (2024) (an HSIC specialization of their Theorem 12), which we adapt to the one-sided encoder class arising in FRHSIC.

Proposition SM6.2 (Uniform concentration of $\widehat{\mathrm{HSIC}}$ over an encoder class). *Let $\mathcal{H}$ be a class of measurable encoders $h : \mathcal{X} \to \mathcal{Z}$. Suppose the kernels $k_{\mathcal{Z}}, k_{\mathcal{S}}$ satisfy Assumption 20 of Ni and Huo (2024): they are bounded with $\sup_z k_{\mathcal{Z}}(z, z) \le \nu_{\mathcal{Z}}$ and $\sup_s k_{\mathcal{S}}(s, s) \le \nu_{\mathcal{S}}$, and the feature maps $z \mapsto k_{\mathcal{Z}}(z, \cdot)$ and $s \mapsto k_{\mathcal{S}}(s, \cdot)$ are Lipschitz with constants $l_{\mathcal{Z}}, l_{\mathcal{S}} > 0$, respectively. Let*

$$\hat{\mathcal{G}}_n(\mathcal{H}) := \mathbb{E}_\xi\left[\sup_{h\in\mathcal{H}} \frac{1}{n}\sum_{i=1}^n \langle \xi_i, h(X_i)\rangle \,\middle|\, \{X_i\}_{i=1}^n\right]$$

denote the empirical Gaussian complexity of $\mathcal{H}$ (Ni and Huo, 2024, Equation 3.5), where $\xi_1, \dots, \xi_n$ are i.i.d. $\mathcal{N}(0, I_{\dim(\mathcal{Z})})$ random vectors. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\begin{aligned}\text{(SM6.1)}\quad \sup_{h\in\mathcal{H}} \left|\widehat{\mathrm{HSIC}}(h(X), S) - \mathrm{HSIC}(h(X), S)\right| \;\le\; & 8\sqrt{2}\,\nu_{\mathcal{Z}}\nu_{\mathcal{S}}\sqrt{\tfrac{\log(2/\delta)}{n}} + \frac{4\,\nu_{\mathcal{Z}}\nu_{\mathcal{S}}}{n} \\ & + 48\sqrt{\pi}\,\max\{\nu_{\mathcal{Z}}\, l_{\mathcal{S}}, \nu_{\mathcal{S}}\, l_{\mathcal{Z}}\}\,\mathbb{E}[\hat{\mathcal{G}}_n(\mathcal{H})].\end{aligned}$$

Proof. Apply Corollary 23 of Ni and Huo (2024) to the two-sample statistic γ_k^2 on the product RKHS with reproducing kernel $k\big((z, s), (z', s')\big) = k_{\mathcal{Z}}(z, z')\, k_{\mathcal{S}}(s, s')$, identified with HSIC via $\mathrm{HSIC}(Z, S) = \gamma_k^2((Z, S), (Z', S'))$ for $(Z', S') \sim P_Z \otimes P_S$ (Ni and Huo, 2024, Equation 3). The bivariate function class $\mathcal{H} \times \{\mathrm{id}_{\mathcal{S}}\}$, where $\mathrm{id}_{\mathcal{S}}$ is the identity on $\mathcal{S}$, satisfies $\mathbb{E}[\mathcal{G}((\mathcal{H} \times \{\mathrm{id}_{\mathcal{S}}\})(\mathbf{X}, \mathbf{S}))] \le \mathbb{E}[\hat{\mathcal{G}}_n(\mathcal{H})]$ by Lemma 19 of Ni and Huo (2024), since the second-coordinate class is a singleton: under the no-absolute-value convention adopted by Ni and Huo (2024, Section 3.1), $\mathcal{G}(\{\mathrm{id}_{\mathcal{S}}\}(\mathbf{S})) = \mathbb{E}_\xi\big[\frac{1}{n}\sum_i \langle \xi_i, S_i\rangle\big] = 0$ by symmetry of the Gaussian distribution. Substituting into the HSIC bound of Corollary 23 yields (SM6.1). ∎

Corollary SM6.3 (Train-population HSIC bridge). *Under the assumptions of Proposition SM6.2, for any encoder $\hat{h} \in \mathcal{H}$ obtained from n i.i.d. training samples and any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\text{(SM6.2)}\qquad \mathrm{HSIC}(\hat{h}(X), S) \;\le\; \widehat{\mathrm{HSIC}}(\hat{h}(X), S) + \varepsilon_n(\delta),$$

where

$$\varepsilon_n(\delta) := 8\sqrt{2}\,\nu_{\mathcal{Z}}\nu_{\mathcal{S}}\sqrt{\frac{\log(2/\delta)}{n}} + \frac{4\,\nu_{\mathcal{Z}}\nu_{\mathcal{S}}}{n} + 48\sqrt{\pi}\,\max\{\nu_{\mathcal{Z}} l_{\mathcal{S}}, \nu_{\mathcal{S}} l_{\mathcal{Z}}\}\,\mathbb{E}[\hat{\mathcal{G}}_n(\mathcal{H})]$$

is the uniform-concentration penalty from Proposition SM6.2.

Proof. By Proposition SM6.2, with probability at least $1-\delta$,

$$\sup_{h\in\mathcal{H}} \left|\widehat{\mathrm{HSIC}}(h(X),S) - \mathrm{HSIC}(h(X),S)\right| \le \varepsilon_n(\delta).$$

Specializing to the data-dependent $\hat{h}\in\mathcal{H}$, $\mathrm{HSIC}(\hat{h}(X),S) - \widehat{\mathrm{HSIC}}(\hat{h}(X),S) \le \varepsilon_n(\delta)$, which rearranges to (SM6.2). ∎

Remark SM6.4 (Train-test fairness bridge). Corollary SM6.3 delivers a train-test fairness guarantee when combined with Theorem 3.4 of the main paper applied on an independent test sample. Theorem 3.4 controls the projected empirical second moment $\frac{1}{n}\sum_i (P_{\widetilde{L}}\boldsymbol{\delta}_f)_i^2$, which under the full-rank hypothesis of Corollary 3.5 simplifies to $\frac{1}{n}\sum_i \hat{\delta}_{f,i}^2$ and, by Jensen's inequality, dominates the squared empirical absolute gap for $f=\phi\circ g$ used in any prediction head. Concretely, given a test sample $\{(\widetilde{Z}_i,\widetilde{S}_i)\}_{i=1}^n$ independent of the training sample, Theorem 3.4 gives the deterministic bound

$$\frac{1}{n}\sum_{i=1}^n \left(P_{\widetilde{L}^{\mathrm{test}}}\,\boldsymbol{\delta}_f^{\mathrm{test}}\right)_i^2 \;\le\; \frac{\|f\|^2_{\mathcal{F}_{\mathcal{Z}}}}{\widetilde{\lambda}_S}\,\widehat{\mathrm{HSIC}}^{\mathrm{test}}(\hat{h}(X),S),$$

where $\widehat{\mathrm{HSIC}}^{\mathrm{test}}$, $\widetilde{\lambda}_S$, and the projection $P_{\widetilde{L}^{\mathrm{test}}}$ are computed on the test sample. Applying Proposition SM6.2 once on training and once pointwise on test (the latter being the standard single-encoder $O(n^{-1/2})$ deviation, which is dominated by ε_n),

$$\widehat{\mathrm{HSIC}}^{\mathrm{test}}(\hat{h}(X),S) \;\le\; \mathrm{HSIC}(\hat{h}(X),S)+\varepsilon_n(\delta) \;\le\; \widehat{\mathrm{HSIC}}^{\mathrm{train}}(\hat{h}(X),S)+2\,\varepsilon_n(\delta)$$

with probability at least $1-2\delta$. Hence, with the same probability,

$$\frac{1}{n}\sum_{i=1}^n \left(P_{\widetilde{L}^{\mathrm{test}}}\,\boldsymbol{\delta}_f^{\mathrm{test}}\right)_i^2 \;\le\; \frac{\|f\|^2_{\mathcal{F}_{\mathcal{Z}}}}{\widetilde{\lambda}_S}\left[\widehat{\mathrm{HSIC}}^{\mathrm{train}}(\hat{h}(X),S)+2\,\varepsilon_n(\delta)\right].$$

This is the generalization statement for the FRHSIC fairness surrogate: training-sample HSIC, plus the uniform-concentration penalty, controls the held-out projected empirical second-moment conditional gap up to the test-sample spectral factor $1/\widetilde{\lambda}_S$.

SM7. Additional Experiments.

SM7.1. Transfer Experiment. The key advantage of FRHSIC over Reg-GDP is *transferability*: since HSIC enforces $Z \perp S$, any downstream prediction head built on the learned representation is guaranteed to be fair. To test this empirically, we train representations using each method (at $\lambda=10$), freeze the encoder, and fit four different downstream heads on the frozen representation: linear, 2-layer MLP, random forest, and SVM. We then measure GDP for each head.

Figure SM2 shows the results on all five datasets; we highlight two representative findings. On Crime ($\lambda=100$), FRHSIC attains low GDP variance across heads (std $=0.001$), comparable to or lower than Reg-GDP (0.002) and ADV (0.005). On Adult ($\lambda=10$), FRHSIC reduces cross-head GDP variance from 0.348 (Unfair) to 0.244, a 30% reduction. Notably, Reg-GDP achieves lower variance (0.058) on Adult but at the cost of collapsing accuracy to 0.758 (Table 2 of the main paper). Among methods that preserve accuracy, FRHSIC provides the most consistent fairness across downstream heads. The MI results in Table 2 of the main paper provide complementary evidence: MI measures the total information about S in Z, bounding

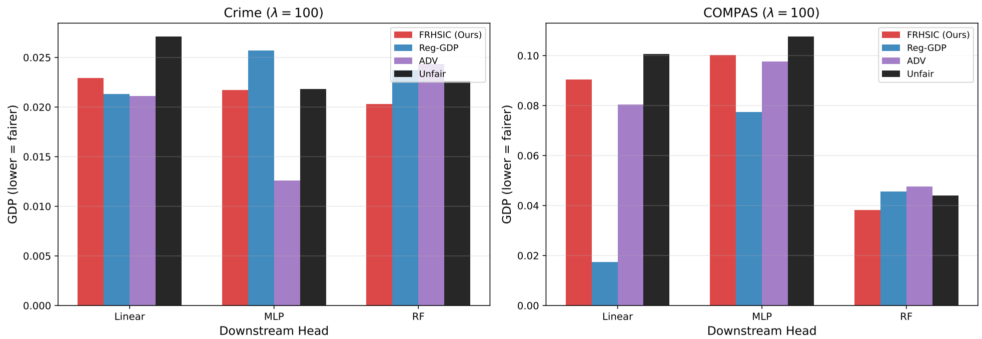

Fig. SM2. *Transfer experiment: GDP of different downstream prediction heads trained on frozen representations. In these experiments FRHSIC maintains comparably low GDP across heads, while Reg-GDP and FREM exhibit more head-dependent variation.*

the discrimination achievable by *any* downstream head (including adversarial ones not tested here). FRHSIC's low MI (0.008 on Adult, 0.015 on Crime) thus guarantees fairness even for worst-case downstream use, while Reg-GDP's higher MI (0.017 on Adult, 0.056 on Crime) leaves room for adversarial exploitation despite its GDP = 0 for the training head.

SM7.2. Empirical Tightness of the Theorem 3.4 Bound. We empirically evaluate the tightness of the bound in Theorem 3.4 of the main paper on two real datasets (Adult and Crime; we omit COMPAS because age in this dataset takes only a small number of distinct integer values, so the centered Gram matrix $\widetilde{L}$ on S is near-low-rank and $\hat{\lambda}_S$ is numerically unstable). For each dataset and each regularization strength λ, we compute (i) the empirical RHS $\|f\|^2/\hat{\lambda}_S \cdot \widehat{\mathrm{HSIC}}(Z, S)$ on the test split, where $\hat{\lambda}_S$ is the smallest positive eigenvalue of $n^{-1}\widetilde{L}$, with $\widetilde{L} = HLH$ the centered Gram matrix on S (eigenvalues below $10^{-6}\,\hat{\lambda}_{\max}$ are treated as numerical zero when identifying the smallest positive eigenvalue, mirroring standard practice for truncated/regularized kernel inverses); and (ii) the empirical LHS $\frac{1}{n}\sum_i \hat{\delta}_{f,i}^2$, averaged over RKHS test functions $f(\cdot) = k_{\mathcal{Z}}(z_0, \cdot)$ for randomly sampled anchors z_0. We plot both as a function of λ on log–log axes.

Figure SM3 shows the results. The bound is valid: the empirical RHS upper-bounds the LHS at every λ on both datasets. The vertical gap between the two curves can span several orders of magnitude and reflects the well-known looseness of HSIC-based bounds when $\hat{\lambda}_S$ is small (the multiplier $1/\hat{\lambda}_S$ is large because $\hat{\lambda}_S$ scales as a small fraction of the largest eigenvalue of $n^{-1}\widetilde{L}$ for the Gaussian kernel at the median-heuristic bandwidth). Crucially, the empirical bound remains finite and non-vacuous — in sharp contrast with the population-level version, where the analogous constant collapses to zero for Gaussian kernels on continuous S and the bound becomes vacuous — so the empirical reformulation of Theorem 3.4 is quantitatively useful as a tractable surrogate.

Bandwidth scaling of $\hat{\lambda}_S$. To validate the polynomial-scaling claim in Remark 3.8 of the main paper, we additionally sweep the bandwidth $\sigma \in \{0.1, 0.3, 1.0, 3.0, 10.0\}$ and record $\hat{\lambda}_S$ for Adult and Crime at $n = 1500$ (Figure SM4). On log–log axes the curves are approximately linear, with negative slopes; the magnitudes are within a factor of two of $d_s = 1$, and the negative sign confirms that the centered Gram matrix on S approaches rank-one as σ grows. This empirical polynomial dependence

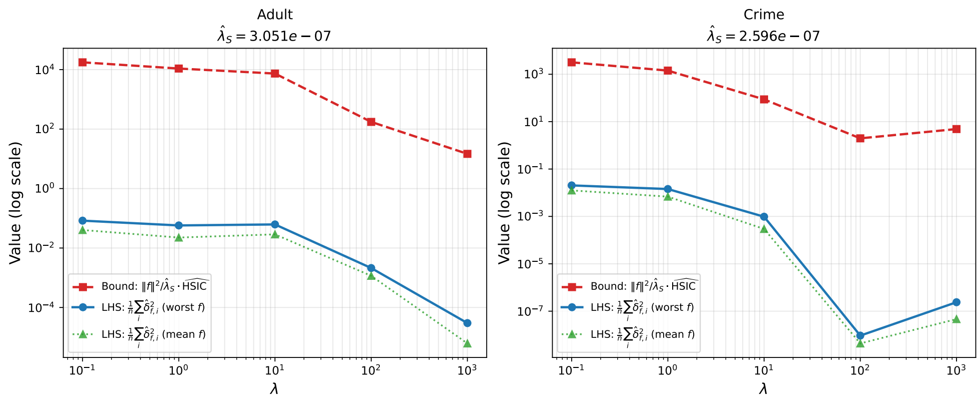

FIG. SM3. *Empirical tightness of the bound from Theorem 3.4 of the main paper. Solid line: empirical LHS* $\frac{1}{n}\sum_i \hat{\delta}^2_{f,i}$. *Dashed line: empirical RHS* $\|f\|^2/\hat{\lambda}_S \cdot \widehat{\mathrm{HSIC}}$. *Validity: the RHS upper-bounds the LHS at every* λ. *The vertical gap can span several orders of magnitude and reflects the looseness of HSIC-based bounds when* $\hat{\lambda}_S$ *is small; the bound nonetheless remains finite and non-vacuous, in contrast to the population version, where the constant collapses to zero.*

is consistent with Remark 3.8 of the main paper.

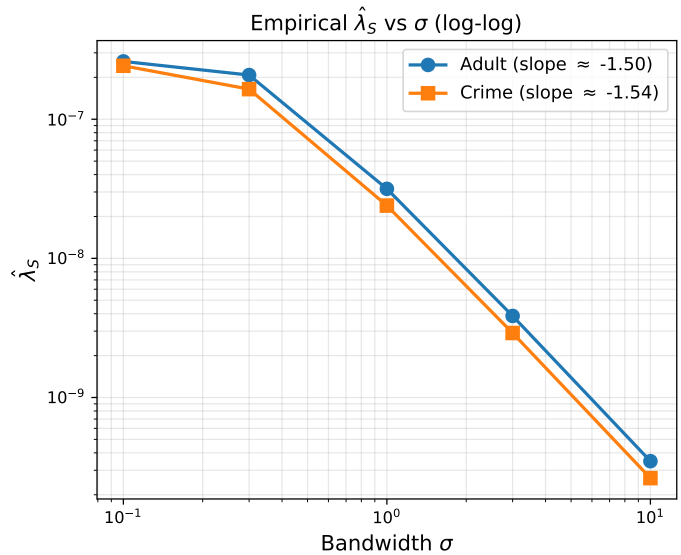

FIG. SM4. *Empirical* $\hat{\lambda}_S$ *as a function of bandwidth* σ *on Adult and Crime,* $n = 1500$. *Log–log axes; the approximately linear scaling (with empirical slopes shown in the legend) supports the polynomial bandwidth dependence noted in Remark 3.8 of the main paper.*

SM7.3. High-Dimensional Sensitive Attributes. A practical consequence of computing on the joint distribution rather than on conditional families is that FRHSIC tolerates increasing $\dim(\mathcal{S})$ gracefully, while methods built on the integral framework $\mathcal{I}_d$ rely internally on a kernel-smoothed conditional weighting $\hat{w}_\gamma(j;i)$ on $\mathcal{S}$ whose statistical accuracy is subject to the curse of dimensionality of nonparametric conditional density estimation. We illustrate the per-iteration computational consequences on the Adult dataset by constructing S as a d-dimensional vector from the continuous Adult features (age, capital gain, capital loss, hours per week, plus a synthetic Gaussian feature for $d = 5$), sweeping $d \in \{1, 2, 3, 4, 5\}$, and comparing the per-epoch training time of FRHSIC and FREM at fixed $\lambda = 10$ and fixed bandwidths ($\sigma_S = 1.0$ for FRHSIC, $\gamma = 0.5$ for FREM); see Figure SM5.

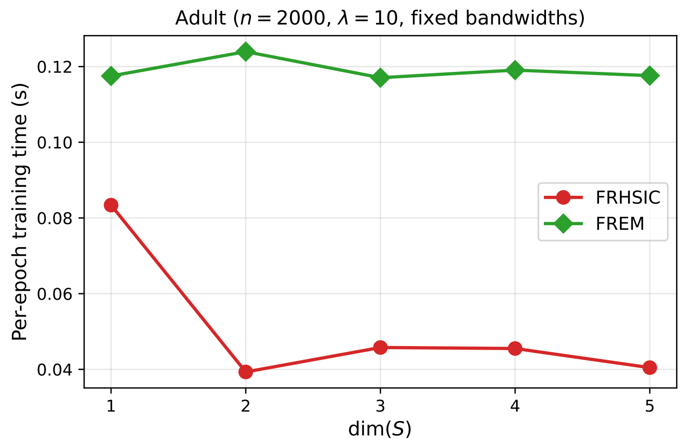

FIG. SM5. *Per-epoch training time on Adult ($n = 2000$) as a function of* $\dim(S)$*, at fixed* $\lambda = 10$ *and fixed kernel bandwidths. FRHSIC's HSIC loss is essentially insensitive to d (the dimension enters only through the $n \times n$ kernel evaluation on S). FREM is consistently 2–3× slower per epoch than FRHSIC across all d; the slowdown is dominated by the kernel-smoothed conditional weighting $\hat{w}_\gamma(j; i)$, which evaluates a d-dimensional Gaussian kernel and an anchor-subsampled MMD on every batch.*

In our wall-clock measurements both curves are roughly flat in d over the range we test, with FREM exhibiting a uniform $\sim$ 2–3× overhead at this scale; the gap grows to roughly 36× on full Adult ($n \approx 3 \times 10^4$, see Section 4.3 of the main paper), where FREM's $O(n^3)$ scaling dominates. At $n = 2000$ with an anchor-subsampled MMD (32 anchors), the per-batch cost is dominated by the $n \times n$ kernel matrix on $\mathcal{Z}$, which is d-independent. The joint-distribution route is therefore primarily a *statistical* (rather than computational) advantage at moderate n: nonparametric conditional weighting on $\mathcal{S}$ inherits the slow rates of d-dimensional density estimation, while the HSIC kernel-matrix computation is d-agnostic. The constant gap we observe nonetheless reflects the per-iteration overhead of constructing conditional weights in $\mathcal{I}_d$-style estimators.

SM7.4. Ablation Studies.

Kernel choice. We evaluate FRHSIC with different kernel functions for $k_\mathcal{Z}$ and $k_\mathcal{S}$: Gaussian (RBF), Laplacian, and inverse multiquadric (IMQ). Table SM4 shows that performance is robust across kernel choices, with the Gaussian kernel performing slightly better overall. This is consistent with the theoretical requirement that the kernels be characteristic, which all three satisfy.

Representation dimensionality. We vary the representation dimension $d_Z \in \{2, 8, 32, 64\}$ with $\lambda = 5$ (Table SM5). All dimensions achieve comparable accuracy (≈ 0.62), but GDP grows with dimensionality, from 0.0001 at $d_Z = 2$ to 0.0034 at $d_Z = 64$. This confirms that higher-dimensional representations make the HSIC penalty less effective per unit of λ. The default $d_Z = 8$ provides a good balance.

TABLE SM4
Ablation over kernel choice (synthetic data, $\lambda = 5$). All kernels achieve comparable performance.

$k_{\mathcal{Z}}$	$k_{\mathcal{S}}$	Accuracy	GDP
Gaussian	Gaussian	0.620	0.0015
Gaussian	Laplacian	0.620	0.0003
Gaussian	IMQ	0.620	0.0006
Laplacian	Gaussian	0.619	0.0091
Laplacian	Laplacian	0.620	0.0001
IMQ	Gaussian	0.620	0.0002
IMQ	IMQ	0.620	0.0027

TABLE SM5
Ablation over representation dimensionality (synthetic data, $\lambda = 5$).

d_Z	Accuracy	GDP
2	0.620	0.0001
8	0.620	0.0001
32	0.618	0.0025
64	0.620	0.0034

SM7.5. Lambda Selection via HSIC Independence Test. We validate the principled regularization strength selection procedure described in Section 3.7 of the main paper. For each dataset, we train FRHSIC over the grid

$$\lambda^{(j)} \in \{0.01, 0.1, 0.5, 1, 5, 10, 50, 100, 500\},$$

evaluate $\widehat{\mathrm{HSIC}}$ on a held-out validation set (15% of data), and apply the gamma-approximation independence test at level $\alpha = 0.05$.

Figure SM6 shows the results. On Crime and COMPAS, the test successfully identifies a transition: as λ increases, the HSIC test statistic falls below the rejection threshold, selecting $\lambda \approx 5$–100 depending on the dataset and random split. At the selected λ, the representation achieves low GDP while preserving prediction performance.

On Adult ($n \approx 30{,}000$), the test rejects independence at all λ values, including $\lambda = 500$ where GDP is near zero. This reflects a well-known property of hypothesis testing: with large samples, the test detects statistically significant but *practically negligible* dependence. In such settings, we recommend supplementing the independence test with a practical significance threshold (e.g., requiring $\widehat{\mathrm{HSIC}} < \epsilon$ for a user-specified ϵ) or using the test on a smaller held-out subsample to reduce power.

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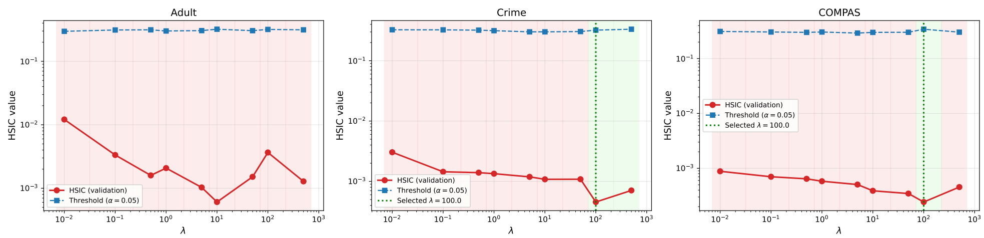

FIG. SM6. *Regularization strength selection via the HSIC independence test (Section 3.7 of the main paper). Red circles: empirical HSIC on the validation set. Blue squares: test threshold at* $\alpha = 0.05$*. Green shading:* λ *values where the test fails to reject independence. The vertical dashed line marks the selected* λ*.*

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